Project Design Phase
Proposed Solution Template

Date	26 June 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No	Parameter	Description
1	**Problem Statement (Problem to be solved)**	Banks and financial institutions face challenges in evaluating loan applications manually, which is time-consuming, error-prone, and may lead to incorrect approval decisions. There is a need for a faster, more accurate, and data-driven loan eligibility assessment system.
2	**Idea / Solution Description**	Smart Lender is a Machine Learning-powered web application that predicts loan eligibility using applicant information such as income, loan amount, credit history, education, employment status, and property area. The system uses Decision Tree, Random Forest, KNN, and XGBoost algorithms, with XGBoost selected as the best-performing model. The model is integrated into a Flask web application for real-time loan approval prediction.
3	**Novelty / Uniqueness**	The solution combines multiple machine learning algorithms and automatically selects the best-performing model for prediction. It provides instant loan eligibility assessment through a user-friendly web interface, reducing dependence on manual evaluation and improving decision consistency.
4	**Social Impact / Customer Satisfaction**	The system helps loan applicants receive quicker responses and improves transparency in loan decisions. Banks benefit from reduced processing time, improved efficiency, and lower risk of approving high-risk applicants. This leads to better customer satisfaction and enhanced financial inclusion.
5	**Business Model (Revenue Model)**	The solution can be offered as a Software-as-a-Service (SaaS) platform to banks and financial institutions through subscription-based licensing, enterprise deployment, or pay-per-prediction usage models.
6	**Scalability of the Solution**	The application can be deployed on IBM Cloud and scaled to handle thousands of loan applications. Additional machine learning models, external credit verification APIs, and multiple banking systems can be integrated in the future without major architectural changes.